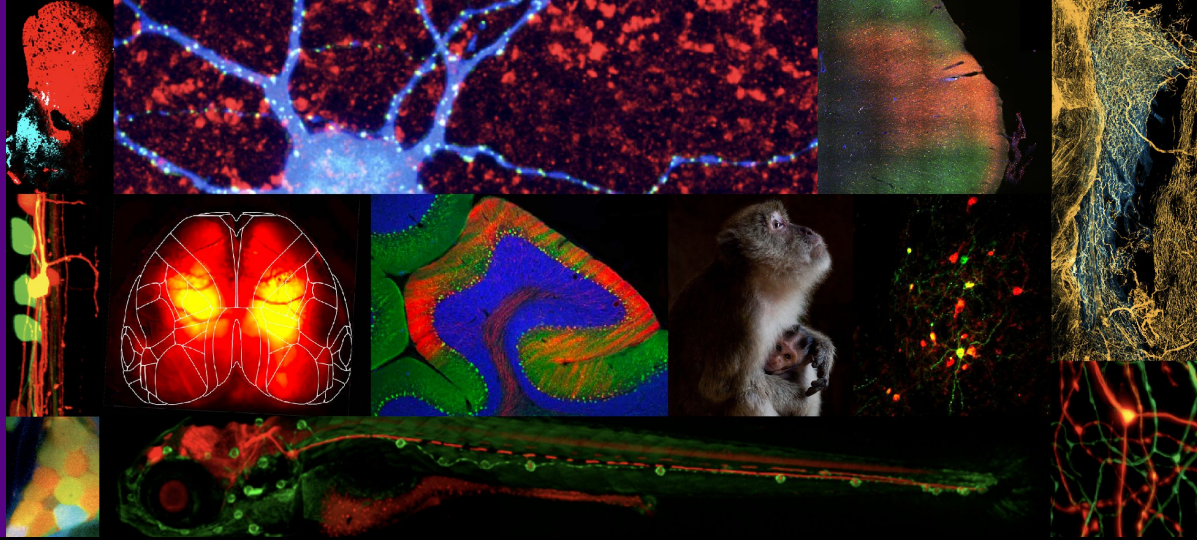


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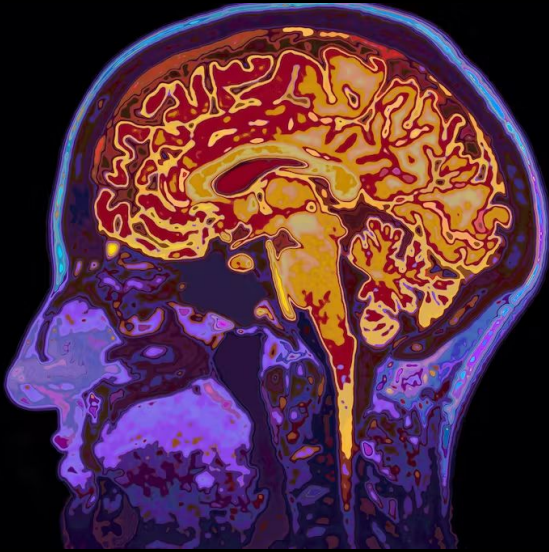


Tate Suarez

11th Grade, Raisbeck Aviation High School

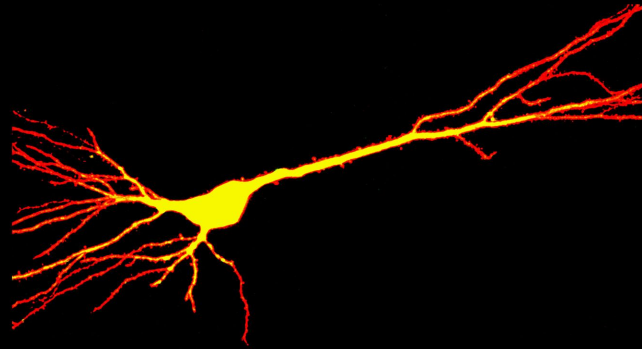
Barria Lab

Research goal of the Barria Lab



Neurodegenerative diseases are a growing problem in the world

The Barria lab studies a specific signaling pathway called **Wnt** that regulates neuronal calcium entry and the communication between neurons.



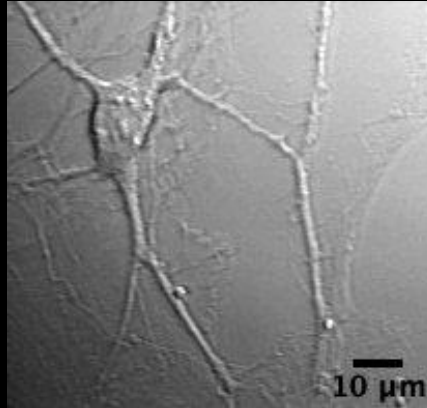
Why is intracellular calcium important?

Intracellular calcium:

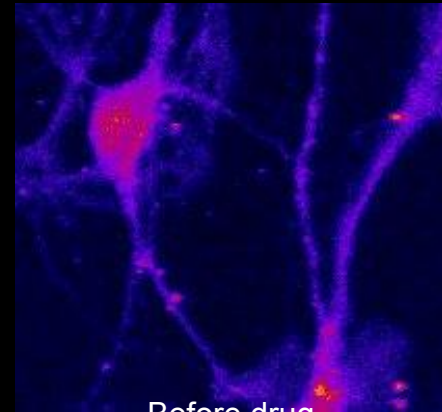
- Muscle contraction
- Neuronal excitability
- Cell migration
- Cell growth

Fluorescent Calcium Imaging

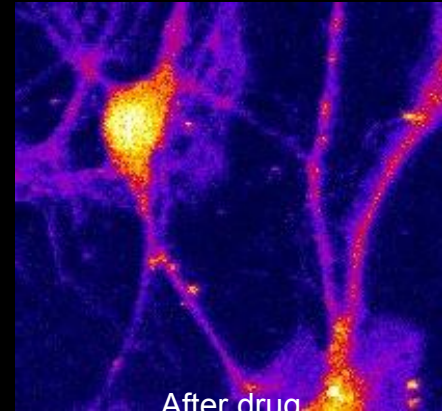
- The purpose of tracking the activity of neurons over time
- To investigate how networks grow or change



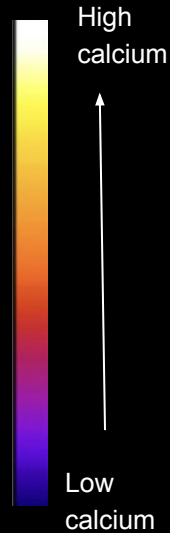
Hippocampal neuron in culture



Before drug stimulation

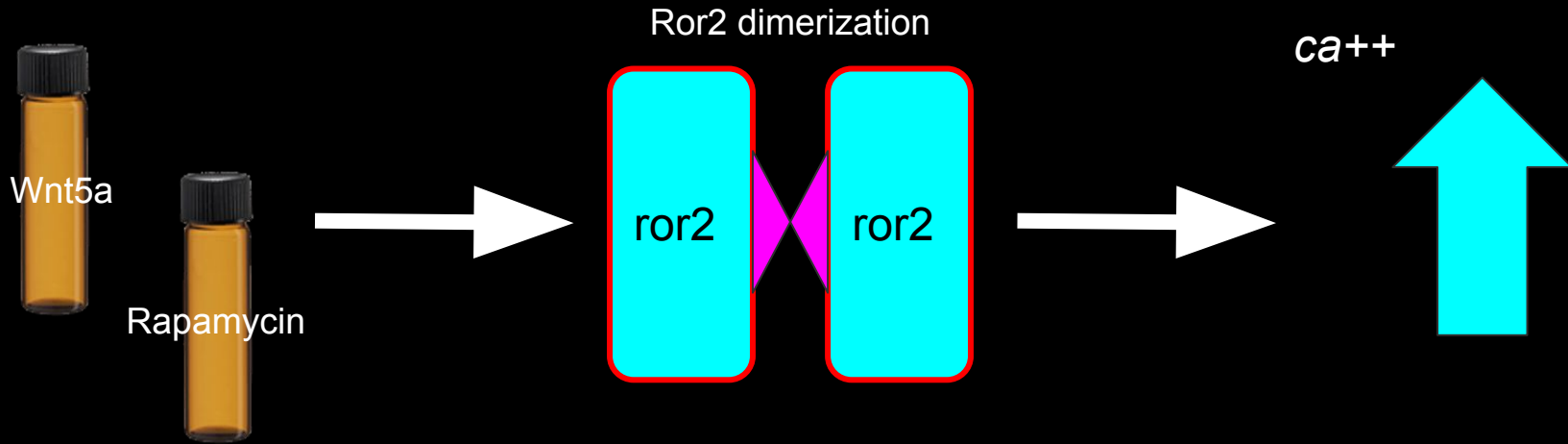


After drug stimulation



Main Question

Is the dimerization of Ror2 able to increase intracellular calcium?



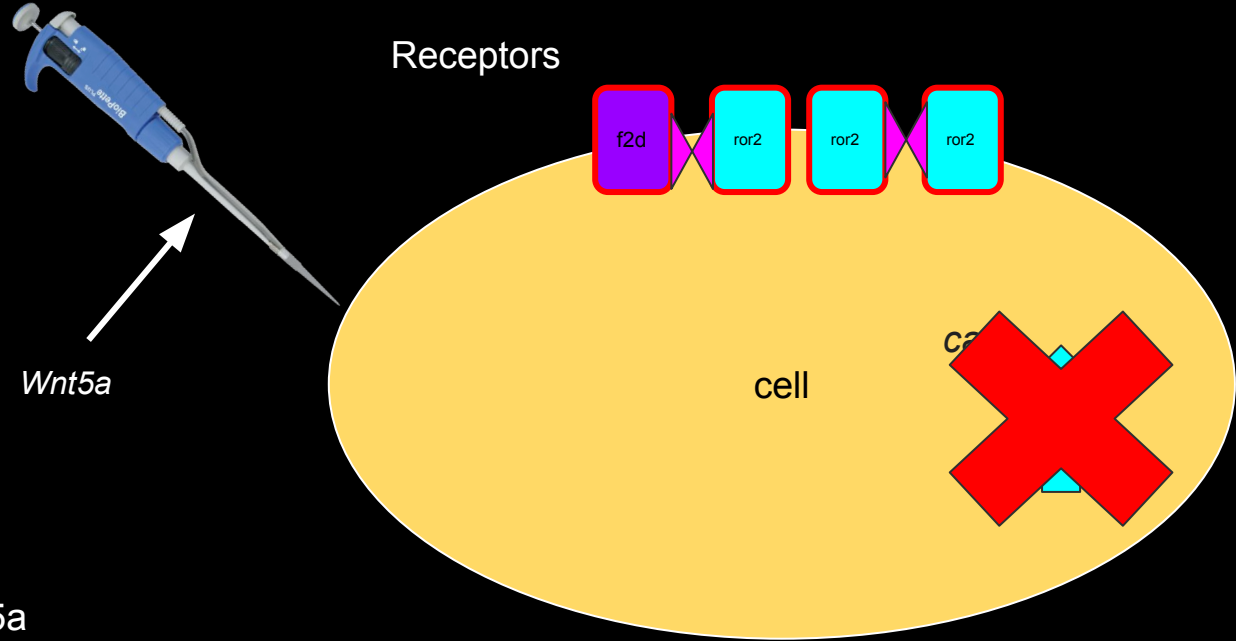
Approach

Intracellular calcium:

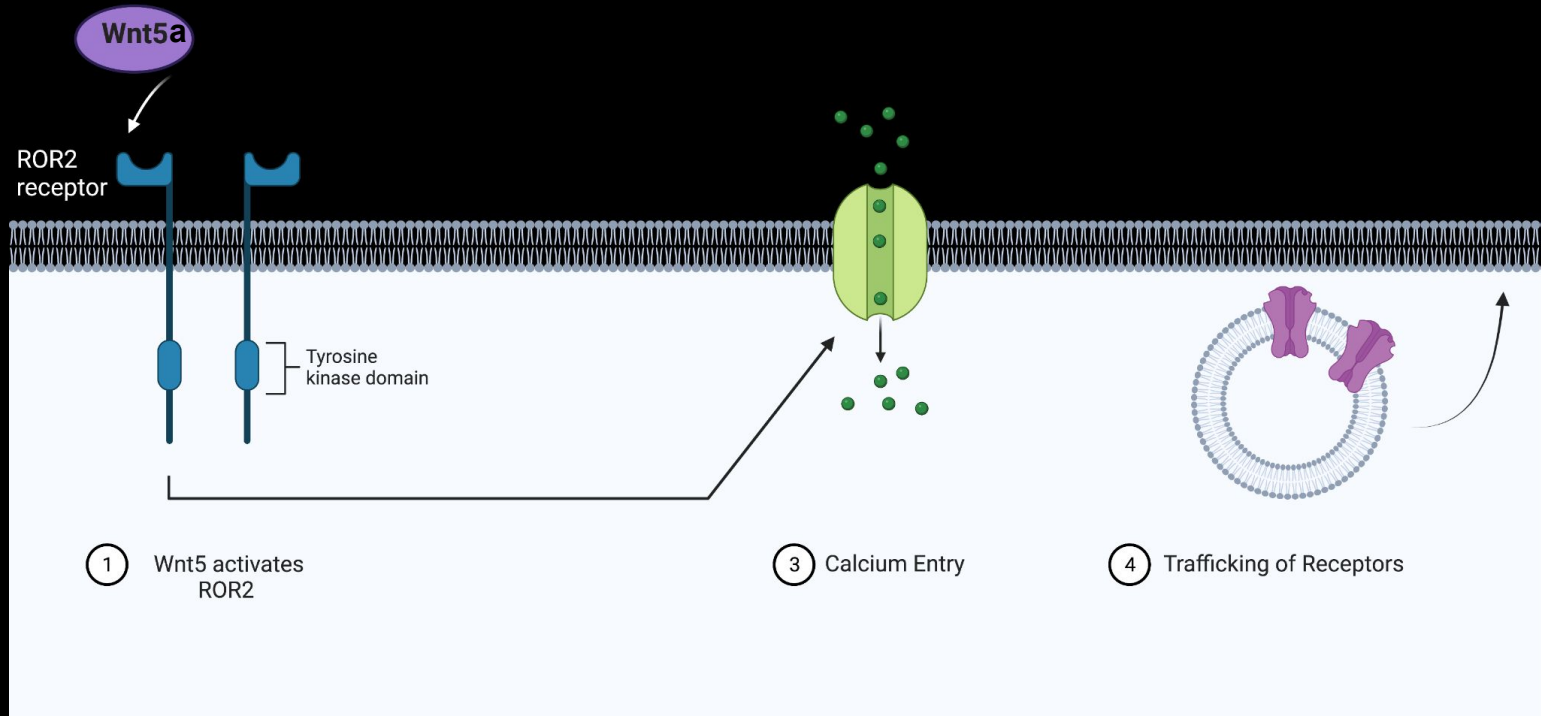
- Muscle contraction
- Neuronal excitability
- Cell migration
- Cell growth

Ror2:

- Signaling receptor
- Detects Wnt, specifically Wnt5a

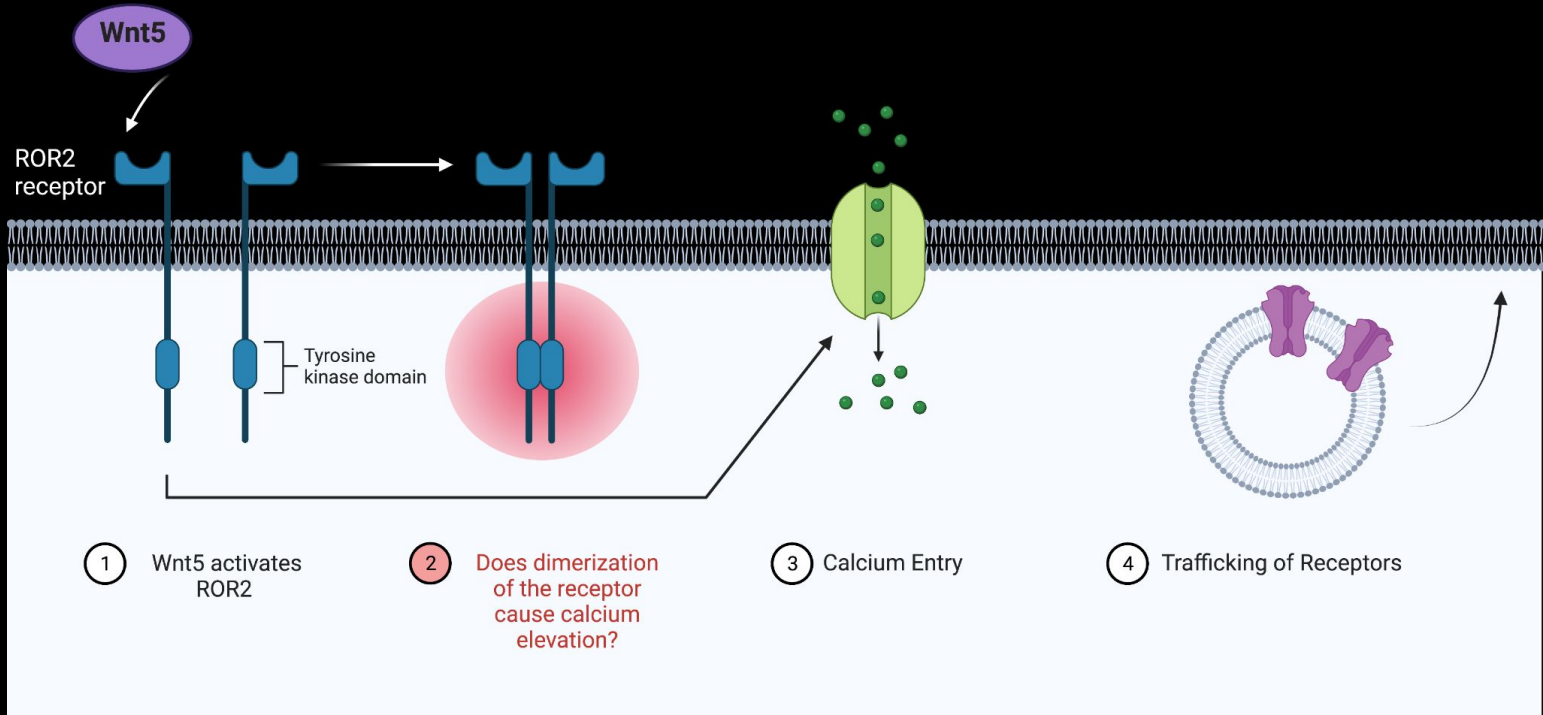


The Wnt signaling pathway

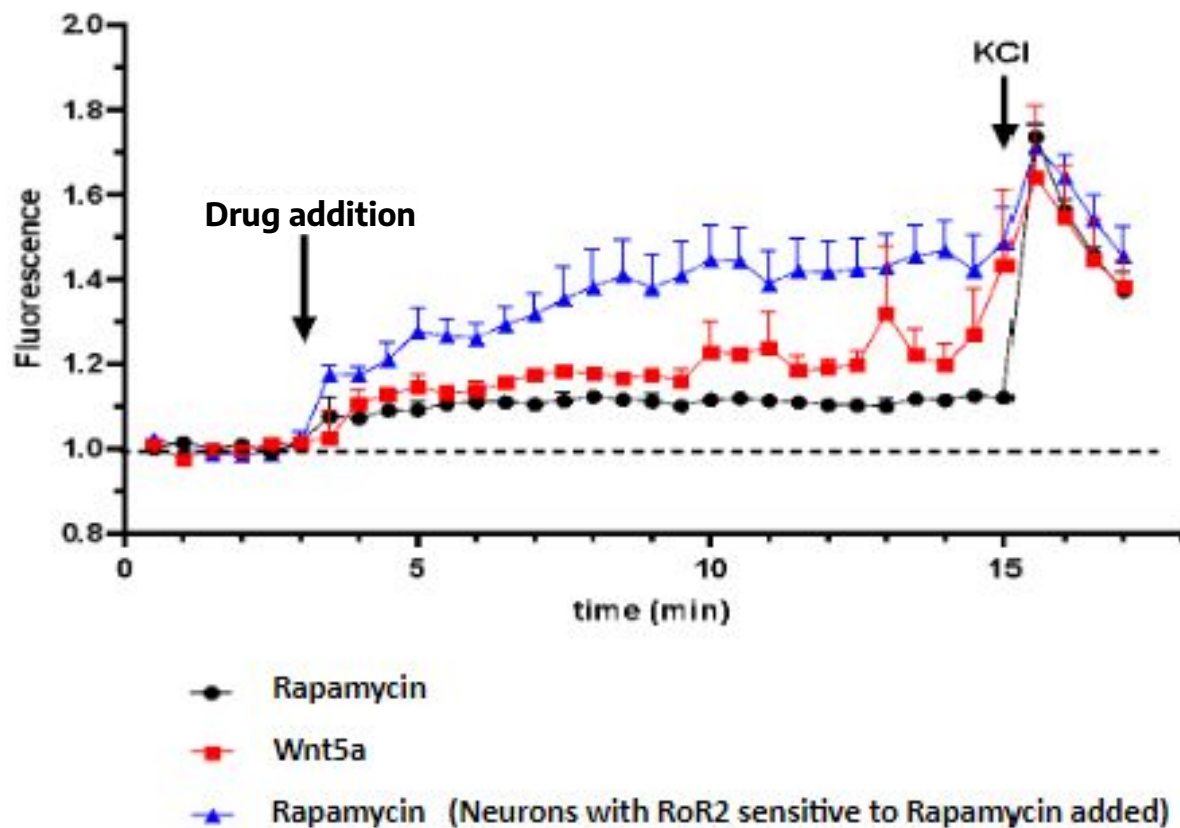


Main question

Is the dimerization of Ror2 able to increase intracellular calcium?

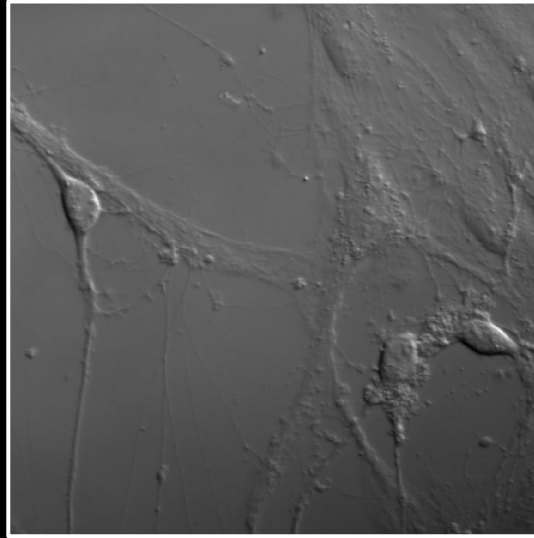
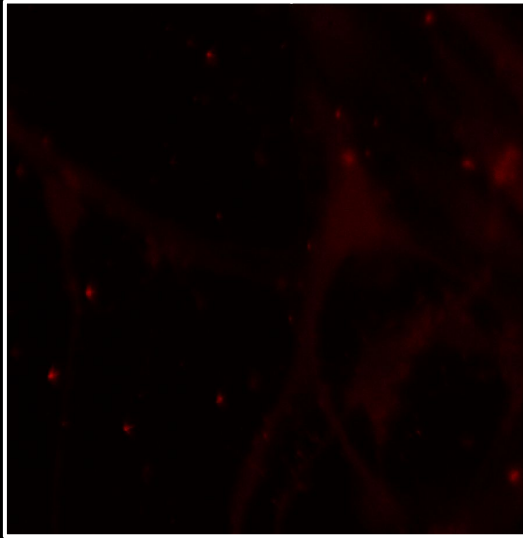


Calcium



Research goal of the Barria Lab

The world needs a solution to the diseases that causes neuronal function to fail.



Dissociated neuronal culture
made from the hippocampus

How can we identify Wnt
participation to reserve
this?

Wnt: /wi-nt/
Intracellular Calcium
Ror2

Acknowledgments

Lizbeth de la Cruz Raul Riquelme-Neculpan Andres Barria

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